Heebert Roshan T

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Education

Kongunadu College of Engineering and Technology — Trichy, Tamil Nadu
B.E. – Electronics and Communication Engineering | *CGPA:* ***8.32*** — *Expected: May 2027*

Lakshmi Matriculation Higher Secondary School — Manapparai, Tamil Nadu
Higher Secondary Certificate (HSC) | ***82%*** — *2022–2023*

Technical Skills

VLSI & Digital Design
- Verilog HDL
- CMOS Logic
- Digital Electronics
- SystemVerilog (Foundational)
- UVM (Foundational)

Programming
- C
- C++
- Python
- Embedded C

EDA Tools & Prototyping
- ModelSim
- Vivado
- MATLAB & Multisim
- Arduino IDE
- Orcad & KiCad
- PCB Design & Fabrication

Embedded & IoT
- NodeMCU (ESP8266)
- Arduino
- ESP32
- Sensor Interfacing
- Blynk Platform
- Relay Control& LoRa

Projects

Self-Adaptive Power-Aware RTL Architecture Using Runtime Toggle Prediction | GitHub 2026
Role: Team Lead
- Proposed the project concept and defined the RTL architecture and implementation approach.
- Developed the Verilog HDL modules and performed functional simulation and waveform analysis using Vivado.
- Coordinated team tasks, integrated modules and managed project development and verification.

VLSI-Based ECG Signal Denoising & Heart Rate Detection | GitHub 2025
Role: Team Lead
- Led RTL architecture development and implementation of ECG signal processing and R-peak detection.
- Developed the Verilog HDL modules and performed functional simulation and waveform analysis using Vivado.

IoT-Based Smart Farm Node – Precision Agriculture System 2025
- Autonomous irrigation system using **NodeMCU (ESP8266)**, soil moisture, DHT11, and PIR sensors with dual-mode (auto + manual) control via **Blynk IoT platform**.

Internship Experience

VLSI Verification Intern — Jul 2026 – Sep 2026
ProV Logic – Hyderabad
- Completed a 2-month VLSI Functional Verification internship focused on **SystemVerilog**, **UVM fundamentals**, and **AMBA-APB** design & verification.

Software & Full Stack Developer Intern — Jul 2024 – Aug 2024
CopterCode – Chennai
- Built production web features using **React.js**; collaborated via **Git** following agile sprint and code review practices.

Achievements & Hackathons

1st Prize – ISTE Tamil Nadu Section Hackathon, Mahendra Engineering College 2026
Designed Affordable Upper-Limb Prosthetic End-Terminal Device – servo-controlled 5-finger grip for partial hand amputees

Special Jury Award (10K) – KreativeGenesis 2026, KGiSL Institutions 2026
Developed AI-Powered Voice-Activated Wearable for Women Safety with offline emergency alert system

Smart India Hackathon (SIH) 2024 – Shortlisted 2024
Government of India National-Level Hackathon – hardware & software innovation tracks

Certifications

- **NPTEL** – Introduction to Industry 4.0 & Industrial Internet of Things (IIoT), Digital Design with Verilog
- **ProV Logic** – VLSI Functional Verification Using SystemVerilog Internship
- **Value Added Courses** – Advanced PCB Design & Core Java, Dept. of ECE, Kongunadu College of Engineering & Technology

Positions of Responsibility

Member & Event Organizer | **IEEE, ISTE, Tamil Mandram, Energy Club** — 2023 – Present
Kongunadu College of Engineering and Technology
- Organized ECE technical workshops; volunteered at **COGNIFY 2026** (24-hr Hackathon) – coordinating participants, managing logistics, and supporting mentors.